COHOMOLOGICAL MILNOR FIBRATIONS OF LANDAU–GINZBURG MODELS

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Abstract. We study the global topology of *fundamentally tame* Landau–Ginzburg (LG) models on complete Kähler manifolds. We first show that the superpotential f is surjective onto $\mathbb{C}$: in the non-proper holomorphic setting this is far from automatic, and here it is deduced from fundamental tameness together with the completeness of M. We then prove that f restricts to a locally trivial fibration over the set of its regular values, which yields a global generalization of the classical Milnor fibration and provides an Ehresmann-type fibration in the non-proper setting. Assuming that f has finitely many critical points, we use a Morse-theoretic argument to show that the relative homology of $\big(M, f^{-1}(c)\big)$, for a regular value c, is concentrated in the middle degree and has rank equal to the global Milnor number μ_f. We further identify the relative homotopy type of the fibers with that of the sublevel sets of f at infinity. Finally, for deformations satisfying a uniform tameness condition on a Stein manifold, we construct the relative cohomology bundles at positive and negative infinity, each equipped with a natural flat Gauss–Manin connection. Their direct sum is called the *cohomological Milnor bundle*.

1. Introduction

Landau–Ginzburg (LG) B-models and tt^*-geometry occupy a central place in mathematical physics and singularity theory, and their local theory is by now classical. Locally one studies a holomorphic function with an isolated singularity at the origin: the basic invariants of the model are supplied by the Milnor fibration together with the Milnor number μ_f [Mil68], by the theory of primitive forms and Frobenius manifolds [Sai81; ST08; Her03], and by the Hodge-theoretic description through the Brieskorn lattice [Sai89], while the topological-field-theoretic background is provided by [CV91; Dub93; Dub96]. The analytic and deformation-theoretic framework underlying the present work was developed by Fan [Fan11; FF21], and the L^2-Hodge theory of the twisted Cauchy–Riemann operators of LG models was established by Li–Wen [LW22] and Wen [Wen16; WF15]; see also [FJR13].

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The passage from the local picture to the global one is, however, not formal, and two difficulties arise. First, it is not evident whether f is surjective onto $\mathbb{C}$. Second, the classical proof of the Milnor fibration over a disk invokes the Ehresmann fibration theorem, whose argument requires the map to be proper; without properness the flow that lifts a vector field from the base need not be complete, and the trivialization argument breaks down. For polynomial functions the corresponding global theory was developed by Broughton [Bro88], where tameness of the polynomial controls the behavior at infinity; see also [AVG12; Bat93; Dim92; Kul98; Lê76] for the singularity-theoretic background and [DFN90] for the homology theory involved.

The purpose of this article is to establish the analogous global structure for *fundamentally tame* LG models on complete Kähler manifolds, within the analytic framework of [Fan11; LW22; Wen16]. Throughout, M is a connected, noncompact, complete Kähler manifold of complex dimension n, and $f : M \to \mathbb{C}$ is a nonconstant holomorphic function. It is fundamental tameness, together with the completeness of M, that replaces properness in the two obstructions above. In the form in which it is used below, fundamental tameness provides a compact subset $K \subset M$ and a constant $C > 0$ such that

$$\|\nabla f(p)\| > C \qquad \text{for all } p \in M \setminus K,$$

so that all critical points of f lie in the compact set K, while the behavior of f away from K is controlled by a uniform gradient bound.

Our first result is a *range theorem* (Theorem 2.1): if (M, f) is a fundamentally tame LG model, then $f(M) = \mathbb{C}$. Its proof is elementary, yet it already exhibits the mechanism that will be used throughout. Writing $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$, one shows that u is unbounded both above and below by following the negative gradient flow of u: as the flow cannot remain in any compact set, completeness of M forces its length to tend to infinity, while the gradient bound forces u to decrease at a uniform rate. The same argument applied to the level sets of u then shows that v is unbounded along $u^{-1}(\alpha)$ for every $\alpha \in \mathbb{R}$, which yields the surjectivity of f.

Our second result is a *fibration theorem* (Proposition 3.1): f restricts to a locally trivial fibration $f^{-1}\big(f(M) \setminus D\big) \to f(M) \setminus D$ over the set of its regular values, where D denotes the set of critical values of f. The argument follows the strategy of the Ehresmann fibration theorem and is inspired by [Bro88, Lemma 4.3]; since f is not assumed to be proper, the lifted vector field is constructed explicitly, and fundamental tameness together with the completeness of M is used to guarantee global existence of its flow. This yields a global generalization of the classical Milnor fibration and provides an Ehresmann-type fibration in the non-proper setting.

Our third result computes the relative homology of the smooth fibers (Theorem 4.1): if (M, f) is a fundamentally tame LG model whose superpotential has finitely many critical points, then for every regular value c of f the relative homology groups $H_*\big(M, f^{-1}(c)\big)$ are concentrated in degree

n, with

$$H_n\big(M, f^{-1}(c)\big) \;\cong\; \mathbb{Z}^{\mu_f},$$

where $\mu_f = \sum_{p \in \mathrm{Crit}(f)} \mu_f(p)$ is the global Milnor number of f. The proof adapts Milnor's classical Morse-theoretic argument [Mil63] to the non-proper, noncompact situation at hand: the auxiliary function $F_a = |f_a - c|^2$ is a nonnegative function whose zero locus is $f^{-1}(c)$ and which, outside a neighborhood of this zero locus, has exactly μ_f nondegenerate critical points, all of Morse index n, while compactness of the sublevel sets is replaced by the completeness of M together with tameness of F_a away from that neighborhood, in the spirit of Broughton [Bro88]. We also clarify the relation between regular fibers and the sublevel sets of f at infinity (Proposition 5.2): for $|\operatorname{Re} c|$ large, the pair $\big(M, f^{-1}(c)\big)$ is homotopy equivalent to the pair formed by M and the corresponding sublevel set, and the common relative homotopy type is independent of c.

Our fourth and last result concerns deformations of the superpotential. For a deformation $F(z,t) = f(z) + \sum_{i=1}^{m} t^i g_i(z)$ over a bounded parameter domain $S \subset \mathbb{C}^m$, and under a uniform tameness assumption (Theorem 5.1), the level set $E_c = \{F = c\}$ is a smooth hypersurface of $M \times S$ for $|\operatorname{Re} c|$ sufficiently large, and the projection $\pi : E_c \to S$ is a locally trivial smooth fibration. If M is Stein, combining this with the homotopy type at infinity yields (Theorem 5.3) the positive and negative relative cohomology bundles $\mathcal{H}^+$ and $\mathcal{H}^-$, each of rank μ and equipped with a natural flat Gauss–Manin connection. We call their direct sum $\mathcal{H}_{\mathrm{tot}} = \mathcal{H}^+ \oplus \mathcal{H}^-$ the *cohomological Milnor bundle* associated with the deformation F. In contrast with the classical Picard–Lefschetz picture, in which the Lefschetz thimbles of a Morse function form a basis of the relative homology, the construction above does not require f_t to be Morse, and both bundles and their flat connections are defined over the whole deformation parameter space.

The article is organized as follows. Section 2 proves the range theorem, together with the auxiliary lemma on maximal integral curves of $-\nabla u$ that the argument requires, and records an application to holomorphic embeddings and polynomial superpotentials. Section 3 establishes the fibration theorem. Section 4 computes the relative homology of the smooth fibers, and Section 5 constructs the cohomological Milnor bundles and relates the regular fibers to the sublevel sets of f at infinity.

2. Range Theorem

As a preparation for our construction, we establish some basic properties of fundamentally tame LG models, beginning with the range theorem for f. This result will later allow us to identify the relative homotopy type of the sublevel sets of f with that of the fibers of f when the real part of f is sufficiently large.

Theorem 2.1. *If (M, f) is a fundamentally tame LG model, then $f(M) = \mathbb{C}$.*

Proof. Let $u = \operatorname{Re} f$ and $v = \operatorname{Im} f$. Then u and v are nonconstant harmonic functions on M.

Step 1: $u(M) = \mathbb{R}$.

Since M is connected, the image $u(M)$ is an interval. To prove $u(M) = \mathbb{R}$, it therefore suffices to show that

$$\inf_{p\in M} u(p) = -\infty \quad \text{and} \quad \sup_{p\in M} u(p) = +\infty. \tag{2.1}$$

We prove the statement for the infimum; the argument for the supremum is analogous.

Assume for contradiction that

$$\alpha_m := \inf_{p\in M} u(p) > -\infty. \tag{2.2}$$

Since f is fundamentally tame, there exist a compact subset $K \subset M$ and a positive constant C such that

$$\|\nabla f(p)\| > C \quad \text{for all } p \in M \setminus K. \tag{2.3}$$

In particular, the critical locus of f is contained in K.

Let

$$\beta_m := \min_{p\in K} u(p). \tag{2.4}$$

By the maximum principle, u does not attain its minimum on M. Hence

$$\alpha_m < \beta_m, \tag{2.5}$$

and consequently

$$(\alpha_m, \beta_m) \subset u(M). \tag{2.6}$$

Moreover,

$$u^{-1}(\alpha_m, \beta_m) \subset M \setminus K. \tag{2.7}$$

Choose $p \in M$ such that $u(p) \in (\alpha_m, \beta_m)$ and consider the negative gradient flow

$$\begin{cases} \phi'(t) = -\nabla u(\phi(t)), \\ \phi(0) = p. \end{cases} \tag{2.8}$$

Let d denote the distance function on M. For $t \geq 0$ we have $u(\phi(t)) \leq u(p) < \beta_m$, hence $\phi(t) \in M \setminus K$. Since $\|\nabla u\| = \frac{1}{2}\|\nabla f\|$, it follows that

$$\|\nabla u(\phi(t))\| > \tfrac{1}{2}C. \tag{2.9}$$

Since

$$u(p) - u(\phi(t)) = \int_0^t -\frac{\mathrm{d}}{\mathrm{d}\tau} u(\phi(\tau))\,\mathrm{d}\tau = \int_0^t \|\nabla u(\phi(\tau))\|^2\,\mathrm{d}\tau, \tag{2.10}$$

we can estimate

$$u(p) - u(\phi(t)) \geq \frac{1}{2}C \int_0^t \|\phi'(\tau)\|\,\mathrm{d}\tau \geq \frac{1}{2}C\, d(\phi(t), p), \tag{2.11}$$

while on the other hand

$$u(p) - u(\phi(t)) \geq \frac{1}{4}C^2 t. \tag{2.12}$$

Let $[0, T)$ be the maximal forward interval of existence of $\phi(t)$.

If $T = +\infty$, then by (2.12),

$$u(p) - u(\phi(t)) \to +\infty \quad \text{as } t \to +\infty, \tag{2.13}$$

which contradicts the assumption that $\alpha_m > -\infty$.

If $T < +\infty$, then by maximality of the interval the curve $\phi(t)$ cannot remain in any compact subset of M. Since M is complete, this implies

$$d(\phi(t), p) \to \infty \quad \text{as } t \to T^-. \tag{2.14}$$

By (2.11) we again obtain

$$u(p) - u(\phi(t)) \to +\infty \quad \text{as } t \to T^-, \tag{2.15}$$

which is a contradiction.

Therefore $\inf_{p\in M} u(p) = -\infty$. Applying the same argument to $-u$ gives $\sup_{p\in M} u(p) = +\infty$, and hence

$$u(M) = \mathbb{R}. \tag{2.16}$$

The argument above in fact yields the following lemma.

Lemma 2.2. *Let M be a complete manifold. If $\gamma : [0, T) \to M$ is a maximal forward integral curve of $-\nabla u$ and $\|\nabla u\| > C$ on $\operatorname{Im}\gamma$ for some $C > 0$, then*

$$u(\operatorname{Im}\gamma) = (-\infty, u(\gamma(0))]. \tag{2.17}$$

Step 2: $v\big(u^{-1}(\alpha)\big) = \mathbb{R}$ for every $\alpha \in \mathbb{R}$, that is,

$$v\big(u^{-1}(\alpha)\big) = \mathbb{R}. \tag{2.18}$$

Since Step 1 shows $u(M) = \mathbb{R}$, the level set $u^{-1}(\alpha)$ is nonempty.

Because f is holomorphic, the Cauchy–Riemann equations imply

$$\nabla v = J(\nabla u), \qquad \langle \nabla u, \nabla v\rangle = 0, \tag{2.19}$$

where J denotes the complex structure. Hence along any integral curve $\gamma(t)$ of $\pm\nabla v$,

$$\frac{\mathrm{d}}{\mathrm{d}t} u(\gamma(t)) = \langle \nabla u, \gamma'(t)\rangle = \pm\langle \nabla u, \nabla v\rangle = 0. \tag{2.20}$$

Therefore every integral curve of $\pm\nabla v$ that starts in $u^{-1}(\alpha)$ remains inside this level set.

Since the critical locus of f is contained in the compact set K, it is a compact complex analytic space. Consider its connected components. By the maximum principle for holomorphic functions on connected complex analytic spaces [GR04, Chapter 5, Section 5], the function f is constant on each connected component, so that on each such component both u and v are constant. Each component is therefore either contained in $u^{-1}(\alpha)$ or disjoint from it.

Suppose that there are k connected components of the critical locus that are contained in $u^{-1}(\alpha)$. If $k = 0$, then $u^{-1}(\alpha)$ contains no critical points of f, so we can choose $C > C_1 > 0$ such that $\|\nabla v\| > C_1$ on $M \setminus K$ and on $u^{-1}(\alpha) \cap K$, hence on $u^{-1}(\alpha)$. Then by Lemma 2.2, $v\big(u^{-1}(\alpha)\big) = \mathbb{R}$. Otherwise, let $k > 0$ and let $S_1, \dots, S_k$ be the connected components of the critical locus that are contained in $u^{-1}(\alpha)$, arranged so that

$$v(S_1) \le \cdots \le v(S_k). \tag{2.21}$$

Since f is a nonconstant holomorphic function, it is an open map. Thus there exists $\epsilon > 0$ such that for each i,

$$D_\epsilon\big(f(S_i)\big) \subset f(M), \tag{2.22}$$

and for $i \neq j$ and $v(S_i) \neq v(S_j)$,

$$D_\epsilon\big(f(S_i)\big) \cap D_\epsilon\big(f(S_j)\big) = \emptyset. \tag{2.23}$$

Hence

$$\big(v(S_i) - \epsilon,\, v(S_i) + \epsilon\big) \subset v\big(u^{-1}(\alpha)\big). \tag{2.24}$$

Choose an open neighborhood $U_\alpha \subset M$ containing $S_1, \dots, S_k$ such that

$$f(U_\alpha) \subset \bigcup_{i=1}^{k} D_{\epsilon'}\big(f(S_i)\big) \tag{2.25}$$

for some $0 < \epsilon' < \epsilon$.

Then

$$(K \setminus U_\alpha) \cap u^{-1}(\alpha) \tag{2.26}$$

is compact, and there exists $\widetilde{C}_\alpha > 0$ such that

$$\|\nabla v\| > \widetilde{C}_\alpha \quad \text{on } u^{-1}(\alpha) \setminus U_\alpha. \tag{2.27}$$

Choose $p_1 \in u^{-1}(\alpha) \setminus U_\alpha$ such that

$$v(p_1) \in \big(v(S_1) - \epsilon,\, v(S_1) - \epsilon'\big). \tag{2.28}$$

Consider the maximal forward integral curve γ_0 of $-\nabla v$ starting at p_1. Since v strictly decreases along this curve, the curve cannot intersect U_α. Let Γ_0 be the image of γ_0. By Lemma 2.2,

$$v(\Gamma_0) = (-\infty, v(p_1)]. \tag{2.29}$$

Set $j_1 = 1$. Choose $q_1 \in u^{-1}(\alpha) \setminus U_\alpha$ such that

$$v(q_1) \in \big(v(S_{j_1}) + \epsilon',\ v(S_{j_1}) + \epsilon\big). \tag{2.30}$$

Consider the maximal forward integral curve γ_1 of ∇v starting at q_{j_1}. If γ_1 does not intersect U_α, then by Lemma 2.2

$$v(\gamma_1) = [v(q_{j_1}), +\infty), \tag{2.31}$$

and the argument terminates.

Otherwise, there exist t_1 and $j_2 > j_1$ such that

$$p_{j_2} := \gamma_1(t_1) \in U_\alpha, \quad v(p_{j_2}) \in \big(v(S_{j_2}) - \epsilon',\ v(S_{j_2}) + \epsilon'\big). \tag{2.32}$$

Let

$$\Gamma_1 = \gamma_1([0, t_1]). \tag{2.33}$$

Repeating the same procedure, we obtain indices

$$1 = j_1 < j_2 < \cdots < j_m \leq k \tag{2.34}$$

for some $1 \leq m \leq k$, points $p_{j_2}, \ldots, p_{j_m} \in U_\alpha$ such that

$$v(p_{j_i}) \in \big(v(S_{j_i}) - \epsilon',\ v(S_{j_i}) + \epsilon'\big), \tag{2.35}$$

points $q_{j_1}, \ldots, q_{j_m} \in u^{-1}(\alpha) \setminus U_\alpha$ such that

$$v(q_{j_i}) \in \big(v(S_{j_i}) + \epsilon',\ v(S_{j_i}) + \epsilon\big), \tag{2.36}$$

and curve segments $\Gamma_1, \ldots, \Gamma_m$, where Γ_k connects q_{j_k} to $p_{j_{k+1}}$ for $k = 1, \ldots, m-1$ and Γ_m is the image of a maximal forward integral curve of ∇v. By Lemma 2.2,

$$v(\Gamma_m) = [v(q_{j_m}), +\infty). \tag{2.37}$$

The sets

$$v(\Gamma_0),\ v(\Gamma_1), \ldots,\ v(\Gamma_m) \tag{2.38}$$

are pairwise disjoint closed intervals whose complement in $\mathbb{R}$ is

$$\big(v(p_{j_1}), v(q_{j_1})\big) \cup \cdots \cup \big(v(p_{j_m}), v(q_{j_m})\big). \tag{2.39}$$

Each of these complementary intervals is contained in one of the open intervals

$$\big(v(S_i) - \epsilon,\ v(S_i) + \epsilon\big), \tag{2.40}$$

which in turn lie in $v\big(u^{-1}(\alpha)\big)$. Therefore

$$v\big(u^{-1}(\alpha)\big) = \mathbb{R}. \tag{2.41}$$

Hence $f(M) = \mathbb{C}$. □

As an application of the range theorem, we record the following corollary concerning holomorphic embeddings and polynomial superpotentials.

Corollary 2.3. *Let (M, f) be a Kählerian LG model, and let $\psi : N \hookrightarrow M$ be a holomorphic embedding. If the induced Kählerian LG model $(N, \psi^* f)$ is fundamentally tame, then $f(M) = \mathbb{C}$. In particular, the following two cases hold:*

1. *If f is a nonconstant polynomial defined on the Euclidean space $\mathbb{C}^N$, then $f(\mathbb{C}^N) = \mathbb{C}$.*
2. *If f is a Laurent polynomial defined on $\big((\mathbb{C}^*)^N, \sum_{i=1}^N |dz_i|^2/|z_i|^2\big)$ having one monomial with positive degree and one monomial with negative degree, then $f\big((\mathbb{C}^*)^N\big) = \mathbb{C}$.*

Proof. Since $(N, \psi^* f)$ is fundamentally tame, the range theorem (Theorem 2.1) applied to this LG model gives $\psi^* f(N) = \mathbb{C}$, and hence $f(M) = \mathbb{C}$. For the polynomial case, choose $v \in \mathbb{C}^N$ such that $z \mapsto f(zv)$ is nonconstant. This one-variable polynomial is surjective by the fundamental theorem of algebra, so $f(\mathbb{C}^N) = \mathbb{C}$. For the Laurent case, choose $\alpha \in (\mathbb{C}^*)^N$ generically so that $h(z) = f(\alpha_1 z, \ldots, \alpha_N z)$ retains a positive and a negative power of z. Such a choice is possible because the coefficient of each total degree is a nonzero Laurent polynomial in α. If $-r < 0$ is the lowest exponent of h, then for every $c \in \mathbb{C}$, $z^r(h(z) - c)$ is a nonconstant polynomial with nonzero constant term. It therefore has a nonzero root, which gives $h(z) = c$ and proves the second case. □

The following class of quiver superpotentials gives examples satisfying the hypothesis of Corollary 2.3 (2).

Example 2.4. Let $Q = (Q_0, Q_1)$ be a finite directed quiver with $|Q_0| \geq 2$, vertex set Q_0, and arrow set Q_1. Attach a nonzero variable $y_v \in \mathbb{C}^*$ to every vertex $v \in Q_0$, and to an arrow $a : u \to v$ associate the character

$$\chi_a = \frac{y_v}{y_u}.$$

The associated quiver superpotential is the Laurent polynomial

$$W_Q = \sum_{a \in Q_1} c_a \, \chi_a = \sum_{a \in Q_1} c_a \, \frac{y_{h(a)}}{y_{t(a)}}, \qquad c_a \in \mathbb{C}^*,$$

where $t(a)$ and $h(a)$ denote the tail and the head of the arrow a. Since the simultaneous rescaling $y_v \mapsto \lambda y_v$ leaves every ratio $y_{h(a)}/y_{t(a)}$ invariant, the natural habitat of W_Q is the algebraic torus

$$T_Q = (\mathbb{C}^*)^{Q_0}/\mathbb{C}^* \simeq (\mathbb{C}^*)^{|Q_0|-1},$$

where one fixes a base vertex $v_0 \in Q_0$ with $y_{v_0} = 1$, endowed with the standard logarithmic metric $\sum_i |dy_i|^2/|y_i|^2$.

The exponent of the arrow $a : u \to v$ is the vector $m_a = e_v - e_u$ in the character lattice $\{m \in \mathbb{Z}^{Q_0} : \sum_{v \in Q_0} m_v = 0\}$. After combining equal characters, the Newton polytope of W_Q is

$$\Delta_Q = \mathrm{Conv}\{m : c_m \neq 0\}, \qquad c_m = \sum_{a \in Q_1 \colon m_a = m} c_a.$$

Assume that Δ_Q is full-dimensional and that $0 \in \mathrm{Int}(\Delta_Q)$. Then the actual support of W_Q contains exponents of both positive and negative total degree in the chosen torus coordinates, so the hypothesis of Corollary 2.3 (2) is satisfied, and we conclude that

$$W_Q(T_Q) = \mathbb{C}.$$

For $n \geq 2$ and $q \in \mathbb{C}^*$, a concrete instance is the Hori–Vafa mirror of $X = \mathbb{P}^{n-1}$, namely

$$W = x_1 + \cdots + x_{n-1} + \frac{q}{x_1 \cdots x_{n-1}} \qquad \text{on } (\mathbb{C}^*)^{n-1},$$

whose monomials x_i have positive degree and whose monomial $\frac{q}{x_1 \cdots x_{n-1}}$ has negative degree; hence $W\big((\mathbb{C}^*)^{n-1}\big) = \mathbb{C}$ by Corollary 2.3 (2).

3. Fibration Theorem

In this section we establish a fibration theorem for the superpotential f. It will be used both in the computation of relative homology and in the comparison between fibers and sublevel sets.

Proposition 3.1. *Let (M, f) be a fundamentally tame LG model. Denote by D the set of critical values of f. Then*

$$f|_{f^{-1}(f(M)\setminus D)} : f^{-1}(f(M) \setminus D) \to f(M) \setminus D \tag{3.1}$$

is a locally trivial fibration.

The argument follows the strategy of the Ehresmann fibration theorem and is inspired by [Bro88, Lemma 4.3]. Since f is a submersion over $f(M) \setminus D$, a trivialization is obtained by lifting a vector field from the base and constructing an isotopy between fibers along its flow. As f is not assumed to be proper, completeness of the lifted flow is not automatic; we therefore construct the lifted vector field explicitly, and use the fundamentally tame condition together with the completeness of M to ensure that its flow exists globally.

Proof. Let $U \subset f(M)$ be a convex subset with $\overline{U} \subset f(M) \setminus D$. Since $f(M) \setminus D$ is open in $\mathbb{C}$, we may choose an open neighborhood V of $\overline{U}$ with $\overline{V} \subset f(M) \setminus D$. Then f is a submersion on $f^{-1}(\overline{V})$.

We look for a vector field $\tilde{X}$ on $f^{-1}(V)$ lifting the constant vector field on the base, that is, satisfying

$$\mathrm{d}f(\tilde{X}) = 1. \tag{3.2}$$

Since f is holomorphic, ∂f is a $(1,0)$-form, and its gradient

$$\nabla f = g^{i\bar{j}} \frac{\partial f}{\partial z^i} \frac{\partial}{\partial \bar{z}^j} \tag{3.3}$$

is a $(0,1)$-vector field. Define

$$\tilde{X} = \frac{1}{\|\nabla f\|^2} \overline{\nabla f}. \tag{3.4}$$

Then $\tilde{X}$ is a smooth $(1,0)$-vector field, and a direct computation gives

$$\tilde{X} f = \frac{1}{\|\nabla f\|^2} \overline{\nabla f}(f) = \frac{1}{\|\nabla f\|^2} \|\nabla f\|^2 = 1. \tag{3.5}$$

Hence $\partial f(\tilde{X}) = 1$, so that $\tilde{X}$ lifts the constant unit vector field on the base.

Because f is fundamentally tame and has no critical point on $f^{-1}(\overline{V})$, the function $\|\nabla f\|$ admits a positive lower bound on $f^{-1}(\overline{V})$. In particular $\tilde{X}$ is bounded.

Choose a smooth function μ on M such that

$$\mu \equiv 1 \ \text{ on } \ f^{-1}(U), \qquad \mu \equiv 0 \ \text{ on } \ M \setminus f^{-1}(V), \qquad 0 \le \mu \le 1, \tag{3.6}$$

and define a global smooth vector field on M by

$$X(p) = \begin{cases} \mu(p)\tilde{X}(p), & p \in f^{-1}(V), \\ 0, & p \notin f^{-1}(V). \end{cases} \tag{3.7}$$

Then X is bounded.

Fix $c_0 \in U$ and, for each $c \in U$, set

$$X_c = (c_0 - c)X, \tag{3.8}$$

denoting the global flow of $2\,\mathrm{Re}\, X_c$ by $\phi_c(p,t)$. Since X_c is bounded and M is complete, this flow is complete.

Then for $p \in f^{-1}(c)$ we have

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\phi_c(p,t)) = (c_0 - c)\mu(\phi_c(p,t)). \tag{3.9}$$

Integrating yields

$$f(\phi_c(p,t)) = c + \int_0^t (c_0 - c)\mu(\phi_c(p,s))\,\mathrm{d}s. \tag{3.10}$$

Since $0 \le \mu \le 1$, for $t \in [0,1]$ we may write

$$f(\phi_c(p,t)) = (1 - \theta(t))c + \theta(t)c_0 \tag{3.11}$$

for some $\theta(t) \in [0,1]$; hence $f(\phi_c(p,t))$ lies on the segment joining c and c_0. Because U is convex, this segment is contained in U, so $\mu = 1$ along the trajectory for $t \in [0,1]$. Therefore

$$f(\phi_c(p,t)) = (1-t)c + tc_0. \tag{3.12}$$

It follows that

$$\phi_c(f^{-1}(c), t) \subset f^{-1}\big((1-t)c + tc_0\big). \tag{3.13}$$

Since the flow is invertible, with inverse given by $t \mapsto -t$, the reverse inclusion also holds. Hence

$$\phi_c(f^{-1}(c), t) = f^{-1}\big((1-t)c + tc_0\big), \tag{3.14}$$

and the induced map

$$\tilde{\phi}_{c,t} : f^{-1}(c) \to f^{-1}\big((1-t)c + tc_0\big) \tag{3.15}$$

is a diffeomorphism.

Define

$$H(p,t) = \phi_{f(p)}(p,t), \qquad p \in f^{-1}(U),\ t \in [0,1]. \tag{3.16}$$

Then

$$\frac{\partial H}{\partial t} = 2\,\mathrm{Re}((c_0 - f(p))X)(H(p,t)), \qquad H(p,0) = p. \tag{3.17}$$

Thus H is smooth and

$$H(f^{-1}(U), 1) = f^{-1}(c_0). \tag{3.18}$$

Hence H defines a deformation retraction of $f^{-1}(U)$ onto $f^{-1}(c_0)$.

Finally define

$$\tilde{H} : f^{-1}(U) \to f^{-1}(c_0) \times U, \qquad \tilde{H}(p) = (H(p,1), f(p)). \tag{3.19}$$

Its inverse is obtained by flowing backwards along the corresponding segment in U. Therefore $\tilde{H}$ is a diffeomorphism, and $f|_{f^{-1}(U)}$ is a trivial fibration over U. □

4. Relative homology of smooth fibers

In this section we compute the relative homology of the regular fibers of f. The resulting description of $H_*(M, f^{-1}(c))$ will be used in the next section to construct the relative cohomology bundle over the deformation parameter space.

Theorem 4.1. *Let (M, f) be a fundamentally tame LG model. Assume that f has finitely many critical points and set $\mu = \mu_f$. Then for any regular value c of f the relative homology groups $H_*(M, f^{-1}(c))$ are concentrated in degree n, and*

$$H_n(M, f^{-1}(c)) \cong \mathbb{Z}^\mu. \tag{4.1}$$

We shall use the following lemma, which is a direct consequence of Lemma B.3 in [Mil68] applied to ∂f.

Lemma 4.2. *Let $V \subset \mathbb{C}^n$ be an open neighborhood of the origin. Let $f : V \to \mathbb{C}$ be a holomorphic function such that the origin is the only critical point of f in V. Let $U \Subset V$ be an open neighborhood of the origin. Then there exists $\epsilon > 0$ such that for almost all tuples $a = (a_1, \ldots, a_n) \in \mathbb{C}^n$ satisfying $\|a\| < \epsilon$, the perturbed function*

$$\tilde{f}(z) = f(z) + a_1 z_1 + \cdots + a_n z_n \tag{4.2}$$

is a Morse function in U. Moreover, the number of critical points of $\tilde{f}$ equals the Milnor number μ_f.

Proof of Theorem 4.1. We follow the procedure in Section 4 of [Bro88] to construct a CW approximation for $(M, f^{-1}(c))$.

Step 1. Let $p_1, \ldots, p_k$ be all critical points of f. Choose pairwise disjoint coordinate charts $(V_i, (z_1^i, \ldots, z_n^i))$ centered at p_i such that $\overline{V_i}$ is compact and $\overline{V_i} \cap f^{-1}(c) = \varnothing$, and open neighborhoods U_i of p_i such that $\overline{U_i} \subset V_i$. Choose smooth functions ϕ^i supported in V_i such that $\phi^i = 1$ on U_i. For

each i, let $a^i = (a^i_1, \dots, a^i_n) \in \mathbb{C}^n$ be a tuple of complex numbers. On V_i define

$$Z_i = (z^i_1 \phi^i, \dots, z^i_n \phi^i), \tag{4.3}$$

and

$$A^i_{a^i} = \left(\sum_{j=1}^{n} a^i_j z^i_j \right) \phi^i = \langle a^i, Z_i \rangle, \qquad A_a = \sum_{i=1}^{k} A^i_{a^i}, \tag{4.4}$$

where each $A^i_{a^i}$ is understood to be zero outside V_i. We then define the perturbed function f_a by

$$f_a = f + A_a. \tag{4.5}$$

Define

$$F_a = |f_a - c|^2, \qquad F = |f - c|^2. \tag{4.6}$$

Clearly F_a coincides with F outside $\bigcup_{i=1}^k V_i$. Let

$$\eta = \min_{\bigcup_i \overline{V_i}} |f - c| > 0.$$

We can restrict the norm of a such that $\|A_a\|_\infty < \eta/2$, thus $F_a \neq 0$ on $\bigcup_{i=1}^k \overline{V_i}$. Then $F_a^{-1}(0) = F^{-1}(0) = f^{-1}(c)$. Let Δ be the open neighborhood of $f^{-1}(c)$ given by the inequality $F_a < \delta^2$ where $0 < \delta < \eta/2$, thus $\overline{\Delta} \subset M \setminus \bigcup_{i=1}^k \overline{V_i}$.

We first prove that F_a is fundamentally tame on $M \setminus \Delta$ and that, if each $\|a^i\|$ is sufficiently small, F_a has no critical point on $M \setminus (\Delta \cup \bigcup_{i=1}^k U_i)$. Since F is real-valued, $\mathrm{d}F = 0$ is equivalent to $\partial F = 0$, and we compute

$$\partial F = \overline{(f - c)}\, \partial f. \tag{4.7}$$

Since f is fundamentally tame, there exists a compact subset $K \subset M$ and a positive constant C such that $\|\partial f\| = \|\nabla f\| > C$ on $M \setminus K$; we may choose K so that $\bigcup_{i=1}^k \overline{V_i} \subset K$. Then on the set $M \setminus (K \cup \Delta)$ we have

$$\|\partial F\| = |f - c|\, \|\partial f\| > \delta C. \tag{4.8}$$

Since F_a coincides with F outside K, the inequality above holds for $\|\partial F_a\|$ as well; hence F_a is fundamentally tame on $M \setminus \Delta$.

We now consider the behavior of F_a on $M \setminus \bigcup_{i=1}^k U_i$. Since $\partial f \neq 0$ and $f \neq c$ on this set, F has no critical point there. Since F_a coincides with F outside $\bigcup_{i=1}^k \overline{V_i}$, it suffices to show that F_a has no critical point on $\overline{V_i \setminus U_i}$ for each i. We argue by estimating $\|\mathrm{d}F_a - \mathrm{d}F\|$ on these sets. Observe

$$F_a = F + 2\,\mathrm{Re}\big((f - c)\overline{A_a}\big) + |A_a|^2. \tag{4.9}$$

Hence

$$\mathrm{d}F_a - \mathrm{d}F = \mathrm{d}\big(2\,\mathrm{Re}((f - c)\overline{A_a})\big) + \mathrm{d}(|A_a|^2). \tag{4.10}$$

We obtain

$$\begin{aligned}
\|\mathrm{d}F_a - \mathrm{d}F\| &= \big\|\mathrm{d}\big(2\,\mathrm{Re}((f-c)\overline{A_a})\big) + \mathrm{d}(|A_a|^2)\big\| \\
&= \big\|2\,\mathrm{Re}(\overline{A_a}\,\mathrm{d}f + (f-c)\,\mathrm{d}\overline{A_a}) + 2\,\mathrm{Re}(\overline{A_a}\,\mathrm{d}A_a)\big\| \\
&\le 2\big(|A_a|\,\|\mathrm{d}f\| + |f-c|\,\|\mathrm{d}A_a\| + |A_a|\,\|\mathrm{d}A_a\|\big) \\
&= 2\big(|\langle a^i, Z_i\rangle|\,\|\mathrm{d}f\| + |f-c|\,\|\langle a^i, \mathrm{d}Z_i\rangle\| \\
&\qquad + |\langle a^i, Z_i\rangle|\,\|\langle a^i, \mathrm{d}Z_i\rangle\|\big) \\
&\le 2\big(\|a^i\|\,\|Z_i\|\,\|\mathrm{d}f\| + |f-c|\,\|a^i\|\,\|\mathrm{d}Z_i\| \\
&\qquad + \|a^i\|\,\|Z_i\|\,\|a^i\|\,\|\mathrm{d}Z_i\|\big).
\end{aligned} \tag{4.11}$$

On the compact set $\overline{V_i \setminus U_i}$ the following functions and forms are bounded:

$$Z_i, \quad \mathrm{d}Z_i, \quad f-c, \quad \partial f. \tag{4.12}$$

Hence there exists a constant $K_i > 0$ such that

$$\|\mathrm{d}F_a - \mathrm{d}F\| \le K_i \|a^i\| \tag{4.13}$$

for $\|a^i\|$ sufficiently small.

Moreover, since ∂f does not vanish on $\overline{V_i \setminus U_i}$, there exists $m_i > 0$ such that

$$\|\mathrm{d}F\| = 2|f-c|\,\|\partial f\| \ge m_i \tag{4.14}$$

on $\overline{V_i \setminus U_i}$.

Choose $\epsilon > 0$ such that

$$K_i \epsilon \le \frac{m_i}{2} \tag{4.15}$$

for every i. Then whenever $\|a^i\| < \epsilon$,

$$\|\mathrm{d}F_a - \mathrm{d}F\| \le \frac{1}{2}\|\mathrm{d}F\| \tag{4.16}$$

on $\overline{V_i \setminus U_i}$.

Second, by Lemma 4.2, we may shrink ϵ so that for almost all a^i with $\|a^i\| < \epsilon$, the function f_a is Morse on U_i and has exactly $\mu_f(p_i)$ critical points in U_i. We fix such an a^i for each i, chosen so that F_a takes distinct values at distinct critical points; this will be used in Step 2. Since f_a is holomorphic on U_i while F_a is real-valued, the critical points of F_a in U_i are determined by

$$\partial F_a = \overline{(f_a - c)}\,\partial f_a = 0. \tag{4.17}$$

Thus they coincide with the critical points of f_a in U_i, and their number equals $\mu_f(p_i)$.

Finally, we compute the Morse index of each critical point of F_a. Since f_a is holomorphic on U_i, the Hessian of F_a in $(z^i, \bar{z}^i)$ coordinates has the form

$$\mathrm{Hess}\, F_a = \begin{pmatrix} \overline{(f_a - c)}\,\partial^2 f_a & (\partial f_a)(\overline{\partial f_a})^T \\ (\overline{\partial f_a})(\partial f_a)^T & (f_a - c)\,\overline{\partial^2 f_a} \end{pmatrix}. \tag{4.18}$$

Since f_a differs from f by linear terms on U_i, their second derivatives coincide:

$$\partial^2 f_a = \partial^2 f. \tag{4.19}$$

At a critical point $p \in U_i$ we have $\partial f_a(p) = 0$, so the off-diagonal blocks of the Hessian vanish. Denote the nonzero constant $\overline{(f_a(p) - c)}$ by λ_p. Then the Hessian of F_a at p has the form

$$\operatorname{Hess} F_a(p) = \begin{pmatrix} \lambda_p\, \partial^2 f(p) & 0 \\ 0 & \overline{\lambda_p\, \partial^2 f(p)} \end{pmatrix}. \tag{4.20}$$

It equals the Hessian at p of the function $\lambda_p f_a + \overline{\lambda_p f_a} = 2\operatorname{Re}(\lambda_p f_a)$. Since $\lambda_p \neq 0$, the functions F_a and $2\operatorname{Re}(f_a)$ have the same Morse index at p. By standard complex Morse theory, the real part of a holomorphic function on a complex n-manifold has Morse index n at a nondegenerate critical point. Hence the Morse index of F_a at p equals n.

Step 2. We construct a CW approximation for $(M, f^{-1}(c))$ by a Morse-theoretic argument in the spirit of Milnor [Mil63] and Broughton [Bro88].

Recall that, by the definition of Δ, $\overline{\Delta} = f_a^{-1}(\overline{D_\delta(c)}) = f^{-1}(\overline{D_\delta(c)})$, where $D_\delta(c)$ denotes the open disk of radius δ centered at c in $\mathbb{C}$. By Proposition 3.1, $f^{-1}(c)$ is a deformation retract of $\overline{\Delta}$; it therefore suffices to understand the homotopy type of $(M, \overline{\Delta})$.

Let

$$M^r = F_a^{-1}[0, r], \qquad M^{s,r} = F_a^{-1}[s, r]. \tag{4.21}$$

Clearly $M^{\delta^2} = \overline{\Delta}$.

As in Theorems 3.1 and 3.2 of [Mil63], we claim:

(1) If $r \geq s \geq \delta^2$ and F_a has no critical point in $M^{s,r}$, then M^s is a deformation retract of M^r.
(2) If p is a critical point of index k with $F_a(p) = s$, then for sufficiently small $\epsilon > 0$, $M^{s+\epsilon}$ has the homotopy type of $M^{s-\epsilon}$ with a k-cell attached.

Then induction on the critical values of F_a shows that M has the homotopy type of $\overline{\Delta}$ with μ n-cells attached.

In Milnor's argument the sets $M^{s,r}$ were assumed compact. As observed by Broughton, the argument remains valid under the weaker assumptions that

(1) each $M^{s,r}$ is complete with respect to the given Riemannian metric;
(2) $\|\nabla F_a\|$ is bounded away from 0 outside a compact neighborhood of the critical points in $M^{s,r}$.

In the present situation M is complete and F_a is fundamentally tame on $M \setminus \Delta$, so the two conditions above are satisfied for every $M^{s,r}$ with $s \geq \delta^2$. Therefore the proofs of Theorems 3.1 and 3.2 in [Mil63] apply without modification, and our claims hold.

We conclude that M has the homotopy type of $f^{-1}(c)$ with μ n-cells attached. Consequently,

$$H_k(M, f^{-1}(c)) = 0 \quad \text{for } k \neq n, \tag{4.22}$$

and

$$H_n(M, f^{-1}(c)) \cong \mathbb{Z}^\mu. \tag{4.23}$$

□

Remark 4.3. When $M = \mathbb{C}^{n+1}$ and f is a tame polynomial in the sense of Broughton, the statement above reduces to a special case of [Bro88, Theorem 1.2], where only regular fibers are considered. Since $\mathbb{C}^{n+1}$ is contractible, the homology of a regular fiber can be recovered from the relative CW approximation of the pair $(M, f^{-1}(c))$ obtained from the Morse-theoretic construction.

In the present setting, however, M is a complete Kähler manifold and need not be contractible. What one obtains is therefore the relative homology of the pair $(M, f^{-1}(c))$ rather than the homology of the fiber itself.

5. Construction of Cohomological Milnor Fibrations

In this section we study deformations of the superpotential f and the fibration structure they induce. Using a fibration theorem analogous to Proposition 3.1, we show that nearby fibers are related by isotopies in M depending smoothly on the deformation parameter. It follows that the relative cohomology groups $H^n(M, f_t^{-1}(c))$ form a smooth vector bundle over the parameter space, equipped with a natural flat connection. To relate this construction to the Lefschetz thimbles of Picard–Lefschetz theory, we also identify the relative homotopy type of the fibers with that of the corresponding sublevel sets at infinity.

Let $S \subset \mathbb{C}^m$ be a bounded open neighborhood of 0. We consider deformations of f of the form

$$F(z,t) = f(z) + \sum_{i=1}^{m} t^i g_i(z), \quad (z,t) \in M \times S, \tag{5.1}$$

where the g_i are holomorphic functions on M and $g_1 = 1$.

For each fixed $t \in S$, denote

$$f_t(z) = F(z,t). \tag{5.2}$$

Theorem 5.1. *Suppose that there exist a compact subset $K \subset M$ and constants $C_i > 0$ for $i = 1, \dots, m$ such that for every $t \in S$,*

$$\|\nabla f_t(z)\| > C_i |g_i(z)| \quad \textit{for all } z \in M \setminus K. \tag{5.3}$$

Then for $c \in \mathbb{C}$ with $|\operatorname{Re} c|$ sufficiently large, the level set

$$E_c = \{(z,t) \in M \times S \mid F(z,t) = c\} \tag{5.4}$$

is a smooth hypersurface of $M \times S$, and the projection

$$\pi : E_c \to S \tag{5.5}$$

is a locally trivial smooth fibration.

Proof. First, we prove that E_c is a smooth hypersurface of $M \times S$.

Taking $i = 1$ in the assumption, we obtain $\|\nabla f_t(z)\| > C_1$ for all $z \in M \backslash K$ and all $t \in S$. Hence all critical points of f_t are contained in K, uniformly for $t \in S$. Moreover, since S is bounded, the values of f_t on K are uniformly bounded for $t \in S$; choose $R > 0$ such that

$$|\operatorname{Re} f_t(z)| \leq R \quad \text{for all } z \in K \text{ and all } t \in S. \tag{5.6}$$

Then for any $c \in \mathbb{C}$ with $|\operatorname{Re} c| > R$, $E_c \cap (K \times S) = \varnothing$. Thus for every $(z, t) \in E_c$ we have

$$\|\nabla_z F(z,t)\| = \|\nabla f_t(z)\| \neq 0. \tag{5.7}$$

Here $\nabla_z F$ denotes the gradient of F with respect to the z-variables only. Hence $\mathrm{d}F(z,t) \neq 0$ on E_c, and E_c is a smooth hypersurface of $M \times S$.

Next, we prove that $\pi : E_c \to S$ is a locally trivial smooth fibration.

Fix a point

$$t_0 = (t_0^1, \ldots, t_0^m) \in S. \tag{5.8}$$

Since $S \subset \mathbb{C}^m$ is open, there exist discs

$$D_i \subset \mathbb{C}, \quad 1 \leq i \leq m, \tag{5.9}$$

each centered at t_0^i, such that

$$D_1 \times \cdots \times D_m \subset S. \tag{5.10}$$

For $1 \leq k \leq m$, denote

$$U_k = D_1 \times \cdots \times D_k \times \{t_0^{k+1}\} \times \cdots \times \{t_0^m\}. \tag{5.11}$$

It suffices to prove that

$$\pi^{-1}(U_m) \to U_m \tag{5.12}$$

is trivial over U_m. We argue by induction on m; the case $m = 0$ is trivial. Assume that for $m - 1$ the projection is trivial over U_{m-1} and that there exists a smooth deformation retraction of $\pi^{-1}(U_{m-1})$ onto $\pi^{-1}(U_0) = \pi^{-1}(t_0)$ whose restriction to $\pi^{-1}(t^1, \ldots, t^{m-1}, t_0^m)$ is an isotopy for every $(t^1, \ldots, t^{m-1}, t_0^m) \in U_{m-1}$.

On E_c, define the vector field

$$X_m = \frac{\partial}{\partial t^m} - \frac{g_m(z)}{\|\nabla_z F(z,t)\|^2} \overline{\nabla_z F(z,t)}. \tag{5.13}$$

We compute

$$\begin{aligned} X_m F(z,t) &= \frac{\partial F}{\partial t^m} - \frac{g_m(z)}{\|\nabla_z F(z,t)\|^2}\,\overline{\nabla_z F(z,t)}\,F \\ &= g_m(z) - \frac{g_m(z)}{\|\nabla_z F(z,t)\|^2}\,\|\nabla_z F(z,t)\|^2 \\ &= 0. \end{aligned} \tag{5.14}$$

Hence X_m is tangent to E_c.

Moreover,

$$\pi_* X_m = \frac{\partial}{\partial t^m}, \tag{5.15}$$

so X_m lifts the constant vector field $\partial/\partial t^m$.

Take the standard Hermitian metric on S. By the assumption, for every $(z,t) \in E_c$ we have

$$\|X_m\| = \left\| \frac{\partial}{\partial t^m} - \frac{g_m(z)}{\|\nabla_z F(z,t)\|^2}\,\overline{\nabla_z F(z,t)} \right\| \leq 1 + \frac{|g_m(z)|}{\|\nabla_z F(z,t)\|} < 1 + \frac{1}{C_m}. \tag{5.16}$$

Hence X_m is smooth and bounded on E_c.

By an argument analogous to that in Proposition 3.1, there exists a smooth deformation retraction of $\pi^{-1}(U_m)$ onto $\pi^{-1}(U_{m-1})$ whose restriction to $\pi^{-1}(t^1,\dots,t^{m-1},t^m)$ is an isotopy for every $(t^1,\dots,t^{m-1},t^m) \in U_m$. By the induction hypothesis, there also exists a smooth deformation retraction of $\pi^{-1}(U_m)$ onto $\pi^{-1}(t_0)$ whose restriction to $\pi^{-1}(t^1,\dots,t^m)$ is an isotopy for every $(t^1,\dots,t^m) \in U_m$. This yields a trivialization of π over U_m. □

We now clarify the relation between regular fibers and sublevel regions of f.

For $a > 0$, define

$$M_-^a := \{p \in M : \mathrm{Re}(f(p)) \leq -a\}, \tag{5.17}$$

$$M_+^a := \{p \in M : \mathrm{Re}(f(p)) \geq a\}. \tag{5.18}$$

Proposition 5.2. *Let (M,f) be a fundamentally tame LG model. There exist constants $R_- > 0$ and $R_+ > 0$ such that:*

(1) For any $a,b \geq R_-$, the pairs (M,M_-^a) and (M,M_-^b) are homotopy equivalent. Moreover, for any $a \geq R_-$ and any $c \in \mathbb{C}$ with $\mathrm{Re}(c) \leq -R_-$, the pair (M,M_-^a) is homotopy equivalent to $(M,f^{-1}(c))$; we denote their common homotopy type by $(M,f^{-\infty})$.

(2) For any $a,b \geq R_+$, the pairs (M,M_+^a) and (M,M_+^b) are homotopy equivalent. Moreover, for any $a \geq R_+$ and any $c \in \mathbb{C}$ with $\mathrm{Re}(c) \geq R_+$, the pair (M,M_+^a) is homotopy equivalent to $(M,f^{-1}(c))$; we denote their common homotopy type by $(M,f^{+\infty})$.

Proof. We prove the negative case; the positive case is analogous.

For $a \geq 0$, set

$$\mathbb{C}_-^a := \{c \in \mathbb{C} : \mathrm{Re}(c) \leq -a\}. \tag{5.19}$$

Then $M_-^a = f^{-1}(\mathbb{C}_-^a)$.

Since f is fundamentally tame, there exists $R_- > 0$ such that $\mathbb{C}_-^{R_-}$ contains no critical value of f. By Theorem 2.1, $\mathbb{C}_-^{R_-}$ is contained in the range of f; by Proposition 3.1, the restriction

$$f : M_-^{R_-} \to \mathbb{C}_-^{R_-} \tag{5.20}$$

is a smooth locally trivial fibration.

For $a \geq R_-$, the half-plane $\mathbb{C}_-^a$ is a deformation retract of $\mathbb{C}_-^{R_-}$. Since the fibration has no critical value over $\mathbb{C}_-^{R_-}$, this deformation retract lifts to a homotopy equivalence of pairs

$$(M, M_-^a) \simeq (M, M_-^{R_-}). \tag{5.21}$$

Similarly, for any c with $\mathrm{Re}(c) \leq -R_-$, the singleton $\{c\}$ is a deformation retract of $\mathbb{C}_-^{R_-}$. Hence

$$(M, f^{-1}(c)) \simeq (M, M_-^{R_-}). \tag{5.22}$$

This completes the proof. □

We now come to the main result of this section. Combining the canonical homotopy type at infinity with the deformation fibration, we obtain the following global statement.

Theorem 5.3. *Assume the hypotheses of Theorem 5.1 and suppose that M is Stein. Let $(M, f_t^{\pm\infty})$ denote the canonical relative homotopy type defined in Proposition 5.2. Then the relative cohomology groups*

$$\mathcal{H}^{\pm} = \coprod_{t \in S} H^n(M, f_t^{\pm\infty}; \mathbb{C}) \tag{5.23}$$

form a smooth vector bundle over S of rank μ, endowed with a natural flat Gauss–Manin connection.

Moreover, the total space

$$\mathcal{H}_{\mathrm{tot}} = \mathcal{H}^+ \oplus \mathcal{H}^- \tag{5.24}$$

is called the cohomological Milnor bundle associated with the deformation F.

Proof. We prove the statement for $\mathcal{H}^+$; the case of $\mathcal{H}^-$ is analogous. By Theorem 5.1, for $c \in \mathbb{C}$ with $\mathrm{Re}\, c$ sufficiently large, the projection

$$\pi : E_c \to S \tag{5.25}$$

is a locally trivial smooth fibration, where

$$E_c = \{(z, t) \in M \times S \mid F(z, t) = c\}. \tag{5.26}$$

By the assumption, we may choose the constant R_+ of Proposition 5.2 uniformly in t, and then choose c so that Proposition 5.2 applies to f_t for all $t \in S$. Hence the pair $(M, f_t^{-1}(c))$ has the homotopy type $(M, f_t^{+\infty})$ for every $t \in S$.

Fix $t_0 \in S$ and let $U \subset S$ be a contractible open neighborhood of t_0. By Theorem 5.1, there exists a smooth trivialization

$$\Phi : \pi^{-1}(U) \longrightarrow \pi^{-1}(t_0) \times U. \tag{5.27}$$

For $t_1, t_2 \in U$, this induces a diffeomorphism

$$\Phi_{t_1,t_2} : \pi^{-1}(t_1) \longrightarrow \pi^{-1}(t_2), \tag{5.28}$$

depending smoothly on t_1, t_2.

Since the projection to M

$$p_t : \pi^{-1}(t) \longrightarrow f_t^{-1}(c) \tag{5.29}$$

is a diffeomorphism for each t, we obtain a smooth family of diffeomorphisms

$$\Psi_{t_1,t_2} := p_{t_2} \circ \Phi_{t_1,t_2} \circ p_{t_1}^{-1} : f_{t_1}^{-1}(c) \longrightarrow f_{t_2}^{-1}(c). \tag{5.30}$$

Because U is contractible, there exists a smooth deformation retraction

$$r : U \times [0,1] \longrightarrow U, \tag{5.31}$$

with $r(t,0) = t$ and $r(t,1) = t_0$. The composition

$$H(z,t,\tau) = \Psi_{t,r(t,\tau)}(z) \tag{5.32}$$

defines a smooth isotopy in M between the level sets $f_t^{-1}(c)$ and $f_{t_0}^{-1}(c)$ for all $t \in U$.

This isotopy induces canonical isomorphisms

$$H^n(M, f_t^{\pm\infty}; \mathbb{C}) \cong H^n(M, f_{t_0}^{\pm\infty}; \mathbb{C}), \tag{5.33}$$

varying smoothly with t. Hence the relative cohomology groups form a local system over S, and therefore determine a smooth vector bundle equipped with the flat Gauss–Manin connection. By Theorem 4.1, the rank of this bundle equals μ. □

Remark 5.4. In classical Picard–Lefschetz theory, when f is a Morse function the Lefschetz thimbles associated with its critical points form a basis of the relative homology

$$H_n\big(M, \{\operatorname{Re} f \ll 0\}\big), \tag{5.34}$$

see for instance [Mil68]. This description relies on the Morse condition and therefore applies only to those deformation parameters for which the function f_t has nondegenerate critical points.

In contrast, the construction above does not require f_t to be Morse. The relative homotopy type at infinity is identified directly with that of a regular fiber, so that the resulting relative cohomology bundle and its flat connection are defined over the whole deformation parameter space.

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